

# Cold-chain telemetry export

Shipment RX-40718 · Vaccines, 2 to 8 °C · Melbourne distribution centre to Royal Hobart Hospital pharmacy

Status

Delivered 30 September 05:40 · 1 excursion, about 90 min above 8.0 °C

Probe A

min 3.6 °C · max 9.1 °C · mean 4.4 °C

Probe B

min 3.4 °C · max 8.4 °C · mean 4.1 °C

Assessment

Within the manufacturer's stability budget; quarantine not required

## Temperature over the journey

Both probes stayed inside the safe band except during the loading-dock delay at Devonport, when the door was held open for 90 minutes. The logger raised alarm TEMP\_HIGH at 20:30 and cleared it at 22:00.

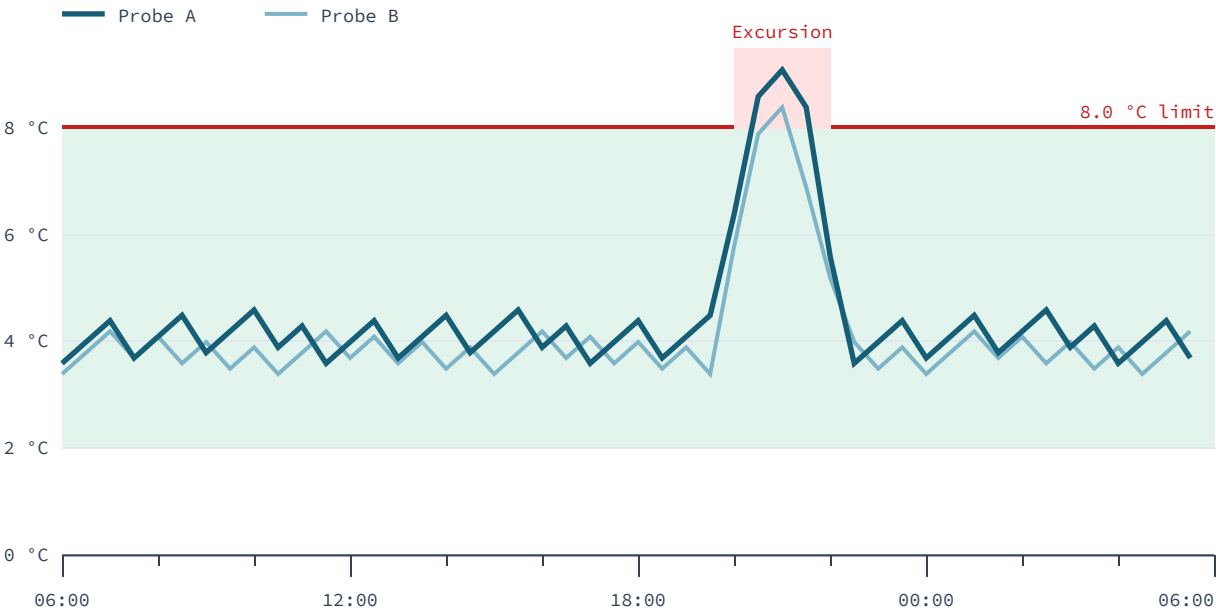


Figure 1. Probe A and probe B against the 2 to 8 °C band; ticks every 2 hours from 06:00

## Excursion record

- 20:00** trailer door opened for transfer to the Bass Strait ferry.
- 21:00** peak of 9.1 °C at probe A, nearest the door.
- 22:00** both probes back below 8.0 °C; alarm cleared.

## Readings

Readings in red exceed the 8.0 °C limit. The logger samples every minute; the table shows the half-hourly values the carrier reports.

Table 1. Half-hourly readings, 06:00 29 September to 05:30 30 September

| Time  | Probe A (°C) | Probe B (°C) | Humidity (%) | Door   | Status    |
|-------|--------------|--------------|--------------|--------|-----------|
| 06:00 | 3.6          | 3.4          | 82           | Closed | In range  |
| 06:30 | 4.0          | 3.8          | 89           | Closed | In range  |
| 07:00 | 4.4          | 4.2          | 87           | Closed | In range  |
| 07:30 | 3.7          | 3.7          | 85           | Closed | In range  |
| 08:00 | 4.1          | 4.1          | 83           | Closed | In range  |
| 08:30 | 3.8          | 3.5          | 80           | Closed | In range  |
| 09:00 | 4.2          | 3.9          | 84           | Closed | In range  |
| 09:30 | 3.9          | 3.6          | 81           | Closed | In range  |
| 10:00 | 4.3          | 4.0          | 85           | Closed | In range  |
| 10:30 | 4.0          | 3.7          | 82           | Closed | In range  |
| 11:00 | 4.4          | 4.1          | 86           | Closed | In range  |
| 11:30 | 4.1          | 3.8          | 83           | Closed | In range  |
| 12:00 | 4.5          | 4.2          | 87           | Closed | In range  |
| 12:30 | 4.2          | 3.9          | 84           | Closed | In range  |
| 13:00 | 4.6          | 4.3          | 88           | Closed | In range  |
| 13:30 | 4.3          | 4.0          | 85           | Closed | In range  |
| 14:00 | 4.7          | 4.4          | 89           | Closed | In range  |
| 14:30 | 4.4          | 4.1          | 86           | Closed | In range  |
| 15:00 | 4.8          | 4.5          | 90           | Closed | In range  |
| 15:30 | 4.5          | 4.2          | 87           | Closed | In range  |
| 16:00 | 4.9          | 4.6          | 91           | Closed | In range  |
| 16:30 | 4.6          | 4.3          | 88           | Closed | In range  |
| 17:00 | 5.0          | 4.7          | 92           | Closed | In range  |
| 17:30 | 4.7          | 4.4          | 89           | Closed | In range  |
| 18:00 | 5.1          | 4.8          | 93           | Closed | In range  |
| 18:30 | 4.8          | 4.5          | 90           | Closed | In range  |
| 19:00 | 5.2          | 4.9          | 94           | Closed | In range  |
| 19:30 | 4.9          | 4.6          | 91           | Closed | In range  |
| 20:00 | 5.3          | 5.0          | 95           | Open   | Excursion |
| 20:30 | 8.5          | 8.0          | 98           | Open   | Excursion |
| 21:00 | 9.1          | 8.4          | 100          | Open   | Excursion |
| 21:30 | 8.8          | 8.1          | 99           | Open   | Excursion |
| 22:00 | 4.5          | 4.2          | 96           | Closed | In range  |
| 22:30 | 4.8          | 4.5          | 97           | Closed | In range  |
| 23:00 | 4.5          | 4.2          | 94           | Closed | In range  |
| 23:30 | 4.9          | 4.6          | 98           | Closed | In range  |
| 00:00 | 4.6          | 4.3          | 95           | Closed | In range  |
| 00:30 | 5.0          | 4.7          | 99           | Closed | In range  |
| 01:00 | 4.7          | 4.4          | 96           | Closed | In range  |
| 01:30 | 5.1          | 4.8          | 100          | Closed | In range  |
| 02:00 | 4.8          | 4.5          | 97           | Closed | In range  |
| 02:30 | 5.2          | 4.9          | 100          | Closed | In range  |
| 03:00 | 4.9          | 4.6          | 98           | Closed | In range  |
| 03:30 | 5.3          | 5.0          | 100          | Closed | In range  |
| 04:00 | 5.0          | 4.7          | 97           | Closed | In range  |
| 04:30 | 5.4          | 5.1          | 100          | Closed | In range  |
| 05:00 | 5.1          | 4.8          | 97           | Closed | In range  |
| 05:30 | 5.5          | 5.2          | 100          | Closed | In range  |

| Time    | Probe A (°C) | Probe B (°C) | Humidity (%) | Door                 | Status    |
|---------|--------------|--------------|--------------|----------------------|-----------|
| 08:30   | 4.5          | 3.6          | 90           | Closed               | In range  |
| 09:00   | 3.8          | 4.0          | 88           | Closed               | In range  |
| 09:30   | 4.2          | 3.5          | 86           | Closed               | In range  |
| 10:00   | 4.6          | 3.9          | 84           | Closed               | In range  |
| 10:30   | 3.9          | 3.4          | 82           | Closed               | In range  |
| 11:00   | 4.3          | 3.8          | 89           | Closed               | In range  |
| 11:30   | 3.6          | 4.2          | 87           | Closed               | In range  |
| 12:00   | 4.0          | 3.7          | 85           | Closed               | In range  |
| 12:30   | 4.4          | 4.1          | 83           | Closed               | In range  |
| 13:00   | 3.7          | 3.6          | 90           | Closed               | In range  |
| 13:30   | 4.1          | 4.0          | 88           | Closed               | In range  |
| 14:00   | 4.5          | 3.5          | 86           | Closed               | In range  |
| 14:30   | 3.8          | 3.9          | 84           | Closed               | In range  |
| 15:00   | 4.2          | 3.4          | 82           | Closed               | In range  |
| 15:30   | 4.6          | 3.8          | 89           | Closed               | In range  |
| 16:00   | 3.9          | 4.2          | 87           | Closed               | In range  |
| 16:30   | 4.3          | 3.7          | 85           | Closed               | In range  |
| 17:00   | 3.6          | 4.1          | 83           | Closed               | In range  |
| 17:30   | 4.0          | 3.6          | 90           | Closed               | In range  |
| 18:00   | 4.4          | 4.0          | 88           | Closed               | In range  |
| 18:30   | 3.7          | 3.5          | 86           | Closed               | In range  |
| 19:00   | 4.1          | 3.9          | 84           | Closed               | In range  |
| 19:30   | 4.5          | 3.4          | 82           | Closed               | In range  |
| 20:00   | 6.4          | 5.8          | 89           | Open                 | In range  |
| 20:30   | 8.6          | 7.9          | 87           | Open                 | Excursion |
| 21:00   | 9.1          | 8.4          | 85           | Open                 | Excursion |
| 21:30   | 8.4          | 6.9          | 83           | Closed               | Excursion |
| 22:00   | 5.6          | 5.2          | 90           | Closed               | In range  |
| 22:30   | 3.6          | 4.0          | 88           | Closed               | In range  |
| 23:00   | 4.0          | 3.5          | 86           | Closed               | In range  |
| 23:30   | 4.4          | 3.9          | 84           | Closed               | In range  |
| 00:00   | 3.7          | 3.4          | 82           | Closed               | In range  |
| 00:30   | 4.1          | 3.8          | 89           | Closed               | In range  |
| 01:00   | 4.5          | 4.2          | 87           | Closed               | In range  |
| 01:30   | 3.8          | 3.7          | 85           | Closed               | In range  |
| 02:00   | 4.2          | 4.1          | 83           | Closed               | In range  |
| 02:30   | 4.6          | 3.6          | 90           | Closed               | In range  |
| 03:00   | 3.9          | 4.0          | 88           | Closed               | In range  |
| 03:30   | 4.3          | 3.5          | 86           | Closed               | In range  |
| 04:00   | 3.6          | 3.9          | 84           | Closed               | In range  |
| 04:30   | 4.0          | 3.4          | 82           | Closed               | In range  |
| 05:00   | 4.4          | 3.8          | 89           | Closed               | In range  |
| 05:30   | 3.7          | 4.2          | 87           | Closed               | In range  |
| Minimum | 3.6          | 3.4          | 82           | Door open 90 min     |           |
| Maximum | 9.1          | 8.4          | 90           | 3 samples over limit |           |